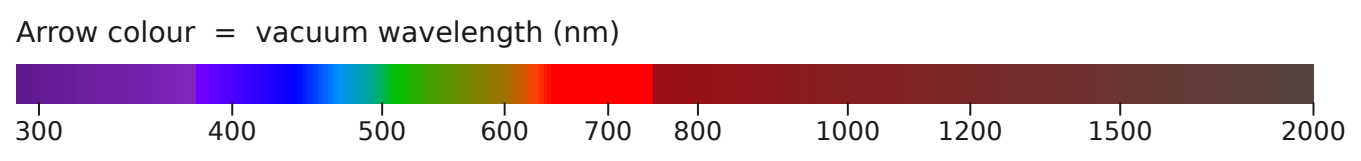
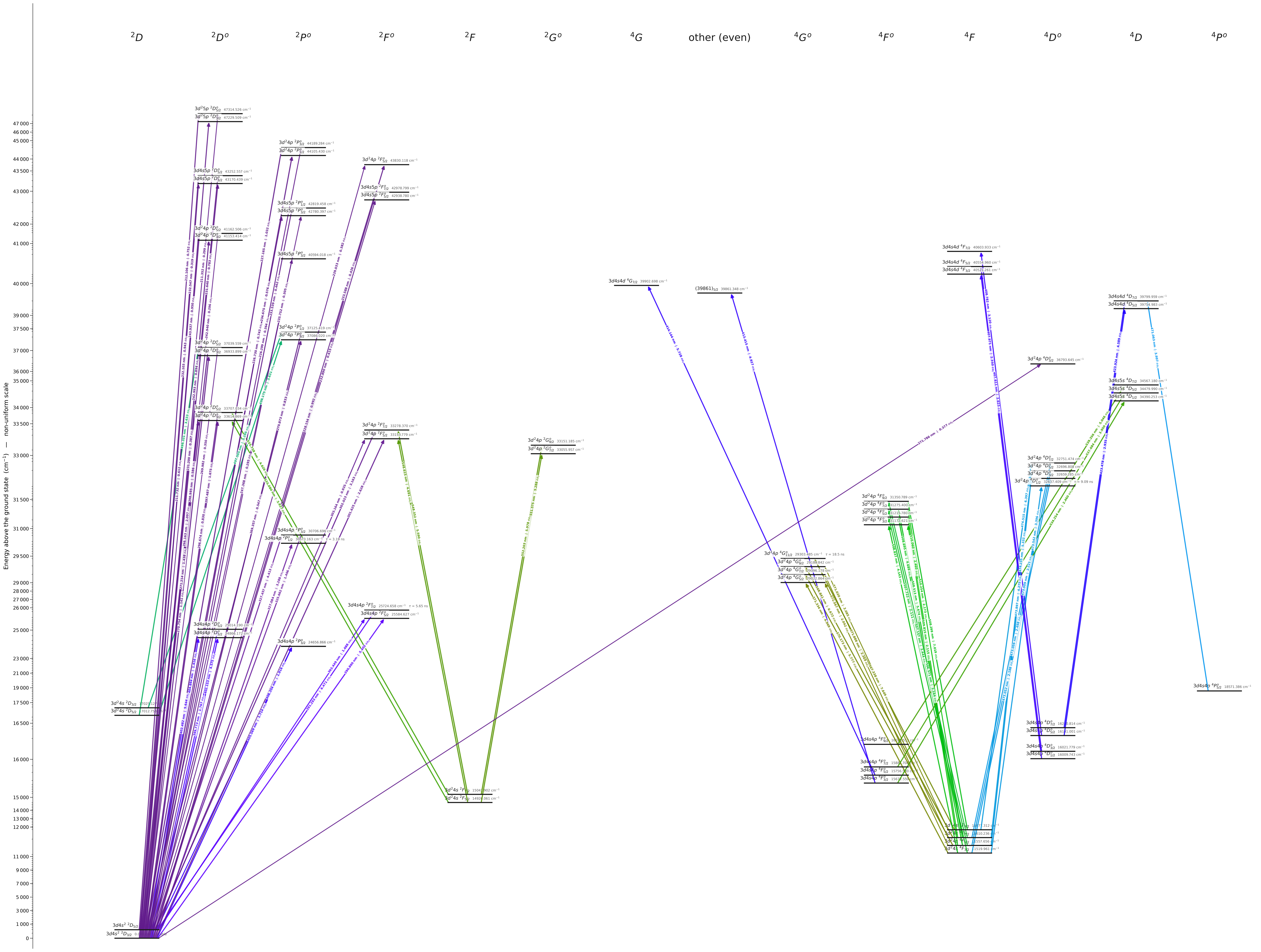
Scandium

Sc (Sc I).   Strongest classified transitions below 2 µm (NIST): 74 levels, 90 lines.

Sc$^+$ ionisation limit   52922.00 cm^{-1} = 6.56149 eV (188.957 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J \| er \| J' \rangle|$ in ea_0
Matrix elements are measured or NIST values;   * = high-accuracy theory,   ≈ = model calculation.
Arrow width grows with the Einstein coefficient A.   Dotted: forbidden (clock, M1, E2).   Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).





^2D
$^2D^o$
$^2P^o$
$^2F^o$
^2F
$^2G^o$
^4G
other (even)
$^4G^o$
$^4F^o$
^4F
$^4D^o$
^4D
$^4P^o$
Energy above the ground state (cm^{-1})   —   non-uniform scale
47 000
46 000
45 000
44 000
43 500
43 000
42 000
41 000
40 000
39 000
37 500
37 000
36 000
35 000
34 000
33 500
33 000
31 500
31 000
29 500
29 000
28 000
27 000
26 000
25 000
23 000
21 000
19 000
17 500
16 500
16 000
15 000
14 000
13 000
12 000
11 000
9 000
7 000
5 000
3 000
1 000
0
$3d^25p\ ^2D^o_{5/2}$ 47314.526 cm^{-1}
$3d^25p\ ^2D^o_{3/2}$ 47229.509 cm^{-1}
$3d4s5p\ ^2D^o_{5/2}$ 43252.557 cm^{-1}
$3d4s5p\ ^2D^o_{3/2}$ 43170.439 cm^{-1}
$3d^24p\ ^2D^o_{5/2}$ 41162.506 cm^{-1}
$3d^24p\ ^2D^o_{3/2}$ 41153.414 cm^{-1}
$3d^24p\ ^2D^o_{5/2}$ 37039.559 cm^{-1}
$3d^24p\ ^2D^o_{3/2}$ 36933.899 cm^{-1}
$3d^24p\ ^2D^o_{5/2}$ 33707.034 cm^{-1}
$3d^24p\ ^2D^o_{3/2}$ 33614.869 cm^{-1}
$3d4s4p\ ^2D^o_{5/2}$ 25014.190 cm^{-1}
$3d4s4p\ ^2D^o_{3/2}$ 24866.172 cm^{-1}
$3d^24p\ ^2P^o_{3/2}$ 44189.284 cm^{-1}
$3d^24p\ ^2P^o_{1/2}$ 44105.430 cm^{-1}
$3d4s5p\ ^2P^o_{1/2}$ 42819.458 cm^{-1}
$3d4s5p\ ^2P^o_{3/2}$ 42780.397 cm^{-1}
$3d4s5p\ ^2P^o_{3/2}$ 40594.018 cm^{-1}
$3d^24p\ ^2P^o_{1/2}$ 37125.419 cm^{-1}
$3d^24p\ ^2P^o_{3/2}$ 37086.020 cm^{-1}
$3d4s4p\ ^2P^o_{3/2}$ 30706.696 cm^{-1}
$3d4s4p\ ^2P^o_{1/2}$ 30573.163 cm^{-1} τ = 3.19 ns
$3d4s4p\ ^2P^o_{3/2}$ 24656.866 cm^{-1}
$3d^24p\ ^2F^o_{5/2}$ 43830.118 cm^{-1}
$3d4s5p\ ^2F^o_{7/2}$ 42978.799 cm^{-1}
$3d4s5p\ ^2F^o_{5/2}$ 42938.780 cm^{-1}
$3d^24p\ ^2F^o_{7/2}$ 33278.370 cm^{-1}
$3d^24p\ ^2F^o_{5/2}$ 33153.779 cm^{-1}
$3d4s4p\ ^2F^o_{7/2}$ 25724.658 cm^{-1} τ = 5.65 ns
$3d4s4p\ ^2F^o_{5/2}$ 25584.627 cm^{-1}
$3d^24s\ ^2D_{3/2}$ 17025.12
$3d^24s\ ^2D_{5/2}$ 17012.75
$3d4s^2\ ^2D_{5/2}$
$3d4s^2\ ^2D_{3/2}$ 0.0
$3d^24s\ ^2F_{7/2}$ 15041.902 cm^{-1}
$3d^24s\ ^2F_{5/2}$ 14926.061 cm^{-1}
$3d^24p\ ^2G^o_{9/2}$ 33151.185 cm^{-1}
$3d^24p\ ^2G^o_{7/2}$ 33055.957 cm^{-1}
$3d4s4d\ ^4G_{7/2}$ 39902.698 cm^{-1}
$(39861)_{5/2}$ 39861.348 cm^{-1}
$3d^24p\ ^4G^o_{11/2}$ 29303.485 cm^{-1} τ = 18.5 ns
$3d^24p\ ^4G^o_{9/2}$ 29189.842 cm^{-1}
$3d^24p\ ^4G^o_{7/2}$ 29096.178 cm^{-1}
$3d^24p\ ^4G^o_{5/2}$ 29022.864 cm^{-1}
$3d^24p\ ^4F^o_{9/2}$ 31350.789 cm^{-1}
$3d^24p\ ^4F^o_{7/2}$ 31275.400 cm^{-1}
$3d^24p\ ^4F^o_{5/2}$ 31215.780 cm^{-1}
$3d^24p\ ^4F^o_{3/2}$ 31172.621 cm^{-1}
$3d4s4p\ ^4F^o_{9/2}$
$3d4s4p\ ^4F^o_{7/2}$
$3d4s4p\ ^4F^o_{5/2}$
$3d4s4p\ ^4F^o_{3/2}$
$3d4s4d\ ^4F_{7/2}$ 40603.933 cm^{-1}
$3d4s4d\ ^4F_{5/2}$ 40554.960 cm^{-1}
$3d4s4d\ ^4F_{3/2}$ 40521.261 cm^{-1}
$3d^24s\ ^4F_{9/2}$
$3d^24s\ ^4F_{7/2}$
$3d^24s\ ^4F_{5/2}$
$3d^24s\ ^4F_{3/2}$
$3d^24p\ ^4D^o_{3/2}$ 36793.645 cm^{-1}
$3d^24p\ ^4D^o_{7/2}$ 32751.474 cm^{-1}
$3d^24p\ ^4D^o_{5/2}$ 32696.808 cm^{-1}
$3d^24p\ ^4D^o_{3/2}$ 32659.285 cm^{-1}
$3d^24p\ ^4D^o_{1/2}$ 32637.409 cm^{-1} τ = 9.09 ns
$3d4s4p\ ^4D^o_{7/2}$
$3d4s4p\ ^4D^o_{5/2}$
$3d4s4p\ ^4D^o_{3/2}$ 16021.779 cm^{-1}
$3d4s4p\ ^4D^o_{1/2}$ 16009.743 cm^{-1}
$3d4s4d\ ^4D_{7/2}$ 39799.959 cm^{-1}
$3d4s4d\ ^4D_{5/2}$ 39754.983 cm^{-1}
$3d4s5s\ ^4D_{7/2}$ 34567.180 cm^{-1}
$3d4s5s\ ^4D_{5/2}$ 34479.990 cm^{-1}
$3d4s5s\ ^4D_{1/2}$ 34390.253 cm^{-1}
$3d4s4p\ ^4P^o_{5/2}$ 18571.386 cm^{-1}
212.106 nm | 0.752 ea_0
232.547 nm | 0.319 ea_0
211.352 nm | 0.299 ea_0
231.640 nm | 0.783 ea_0
243.937 nm | 0.950 ea_0
242.940 nm | 0.296 ea_0
232.103 nm | 0.943 ea_0
242.993 nm | 0.894 ea_0
499.331 nm | 3.633 ea_0
211.732 nm | 0.612 ea_0
269.982 nm | 0.350 ea_0
297.487 nm | 1.675 ea_0
296.674 nm | 0.830 ea_0
271.214 nm | 1.141 ea_0
227.165 nm | 1.032 ea_0
234.675 nm | 0.576 ea_0
233.539 nm | 0.462 ea_0
233.752 nm | 0.305 ea_0
226.730 nm | 0.743 ea_0
226.299 nm | 0.364 ea_0
498.175 nm | 3.631 ea_0
497.505 nm | 3.195 ea_0
535.758 nm | 4.626 ea_0
535.080 nm | 3.927 ea_0
247.368 nm | 0.295 ea_0
269.357 nm | 0.547 ea_0
270.873 nm | 0.833 ea_0
228.154 nm | 0.992 ea_0
229.033 nm | 0.382 ea_0
233.588 nm | 0.426 ea_0
232.890 nm | 0.415 ea_0
303.164 nm | 0.935 ea_0
302.023 nm | 3.163 ea_0
301.625 nm | 2.628 ea_0
548.615 nm | 4.691 ea_0
548.352 nm | 5.590 ea_0
551.575 nm | 5.288 ea_0
552.203 nm | 5.978 ea_0
327.457 nm | 4.413 ea_0
327.084 nm | 3.288 ea_0
325.662 nm | 1.466 ea_0
402.482 nm | 5.644 ea_0
404.894 nm | 1.416 ea_0
399.773 nm | 1.767 ea_0
402.153 nm | 4.575 ea_0
405.569 nm | 1.714 ea_0
408.354 nm | 1.916 ea_0
393.449 nm | 1.688 ea_0
391.293 nm | 6.471 ea_0
390.860 nm | 5.402 ea_0
414.144 nm | 5.728 ea_0
413.415 nm | 4.927 ea_0
271.786 nm | 0.377 ea_0
408.782 nm | 3.120 ea_0
407.971 nm | 2.350 ea_0
407.611 nm | 2.623 ea_0
423.924 nm | 4.589 ea_0
423.478 nm | 2.935 ea_0
471.063 nm | 3.997 ea_0
539.356 nm | 5.046 ea_0
537.684 nm | 3.950 ea_0
534.254 nm | 2.460 ea_0
474.515 nm | 6.397 ea_0
473.897 nm | 2.498 ea_0
473.056 nm | 2.188 ea_0
571.334 nm | 4.969 ea_0
568.841 nm | 6.672 ea_0
571.020 nm | 2.085 ea_0
572.567 nm | 2.042 ea_0
570.173 nm | 5.777 ea_0
571.888 nm | 2.369 ea_0
567.338 nm | 7.649 ea_0
508.837 nm | 4.143 ea_0
508.695 nm | 4.689 ea_0
510.060 nm | 2.460 ea_0
507.722 nm | 2.111 ea_0
508.513 nm | 5.674 ea_0
510.254 nm | 2.173 ea_0
507.157 nm | 2.444 ea_0
509.815 nm | 2.115 ea_0
506.571 nm | 2.164 ea_0
508.299 nm | 7.019 ea_0

Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\lvert\langle J \| er \| J' \rangle\rvert$ (ea₀)	A (s⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
$3d^24s\ ^4F_{9/2}$ – $3d^24p\ ^4G^o_{11/2}$	567.3381	567.1807	528.4194	7.649	5.410e+07	8.612	100	laser-cooling

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

^{45}Sc hyperfine structure (I = 7/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$3d4s^2\ ^2D_{3/2}$	269.5581(9)	26.361(9)	F=2: -1805.39 F=3: -1015.55 F=4: +55.15 F=5: +1421.77
$3d4s^2\ ^2D_{5/2}$	109.0330(9)	-37.372(15)	F=1: -1246.64 F=2: -1017.90 F=3: -678.79 F=4: -234.11 F=5: +309.72 F=6: +944.70
$3d^24s\ ^4F_{3/2}$	-158.512(3)	-5.189(21)	F=2: +1067.18 F=3: +595.35 F=4: -37.22 F=5: -833.49
$3d^24s\ ^4F_{5/2}$	154.0352(21)	-6.45(3)	F=1: -1736.35 F=2: -1426.44 F=3: -962.26 F=4: -344.64 F=5: +425.30 F=6: +1346.20
$3d^24s\ ^4F_{7/2}$	250.0119(21)	-9.08(3)	F=0: -3942.55 F=1: -3691.61 F=2: -3189.92 F=3: -2437.85 F=4: -1435.95 F=5: -184.96 F=6: +1314.18 F=7: +3060.38
$3d^24s\ ^4F_{9/2}$	285.966(9)	-15.5(7)	F=1: -5511.94 F=2: -4937.79 F=3: -4077.04 F=4: -2930.23 F=5: -1498.10 F=6: +218.43 F=7: +2218.26 F=8: +4500.10
$3d^24s\ ^2F_{5/2}$	292(6)	–	F=1: -3281.60 F=2: -2698.21 F=3: -1823.11 F=4: -656.32 F=5: +802.17 F=6: +2552.36
$3d^24s\ ^2F_{7/2}$	-18.0(12)	–	F=0: +283.30 F=1: +265.32 F=2: +229.34 F=3: +175.38 F=4: +103.43 F=5: +13.49 F=6: -94.44 F=7: -220.35
$3d4s4p\ ^4F^o_{3/2}$	-137.6(3)	-27(4)	F=2: +914.54 F=3: +520.78 F=4: -22.01 F=5: -729.10
$3d4s4p\ ^4F^o_{5/2}$	289.6(3)	–	F=1: -3257.99 F=2: -2678.80 F=3: -1810.00 F=4: -651.60 F=5: +796.40 F=6: +2534.00
$3d4s4p\ ^4F^o_{7/2}$	359.5(3)	-39(5)	F=0: -5682.23 F=1: -5318.81 F=2: -4592.75 F=3: -3505.64 F=4: -2059.89 F=5: -258.65 F=6: +1894.08 F=7: +4393.53
$3d4s4p\ ^4D^o_{1/2}$	-510.8(18)	–	F=3: +1149.40 F=4: -893.98
$3d4s4p\ ^4D^o_{3/2}$	342.4(24)	–	F=2: -2310.95 F=3: -1283.86 F=4: +85.59 F=5: +1797.41
$3d4s4p\ ^4F^o_{9/2}$	390.9(3)	-63(14)	F=1: -7554.38 F=2: -6763.49 F=3: -5579.03 F=4: -4003.26 F=5: -2039.20 F=6: +309.39 F=7: +3037.99 F=8: +6141.32
$3d4s4p\ ^4D^o_{5/2}$	306.1(6)	–	F=1: -3443.49 F=2: -2831.32 F=3: -1913.05 F=4: -688.70 F=5: +841.74 F=6: +2678.27
$3d4s4p\ ^4D^o_{7/2}$	457.8(3)	22(12)	F=0: -7198.36 F=1: -6742.81 F=2: -5831.26 F=3: -4462.83 F=4: -2636.16 F=5: -349.48 F=6: +2399.45 F=7: +5613.31
$3d^24s\ ^2D_{5/2}$	552.5(6)	89(11)	F=1: -6168.12 F=2: -5088.53 F=3: -3459.59 F=4: -1269.88 F=5: +1495.89 F=6: +4856.79
$3d^24s\ ^2D_{3/2}$	-288.4(9)	-17(8)	F=2: +1937.71 F=3: +1084.50 F=4: -64.31 F=5: -1518.30
$3d4s4p\ ^4P^o_{5/2}$	587.9(6)	–	F=1: -6613.80 F=2: -5438.01 F=3: -3674.33 F=4: -1322.76 F=5: +1616.71 F=6: +5144.06
$3d4s4p\ ^2P^o_{1/2}$	442(5)	–	F=3: -994.26 F=4: +773.32
$3d4s4p\ ^2P^o_{3/2}$	12.6(9)	-191(13)	F=2: -187.13 F=3: -13.17 F=4: +91.67 F=5: +18.44
$3d4s4p\ ^2D^o_{3/2}$	278(5)	–	F=2: -1875.88 F=3: -1042.15 F=4: +69.48 F=5: +1459.02
$3d4s4p\ ^2D^o_{5/2}$	31(4)	–	F=1: -347.38 F=2: -285.63 F=3: -192.99 F=4: -69.48 F=5: +84.92 F=6: +270.19
$3d4s4p\ ^2F^o_{5/2}$	171.5(15)	–	F=1: -1929.16 F=2: -1586.20 F=3: -1071.76 F=4: -385.83 F=5: +471.57 F=6: +1500.46
$3d4s4p\ ^2F^o_{7/2}$	125(3)	–	F=0: -1964.24 F=1: -1839.53 F=2: -1590.10 F=3: -1215.96 F=4: -717.10 F=5: -93.53 F=6: +654.75 F=7: +1527.74
$3d^24p\ ^4G^o_{5/2}$	279.1(12)	-29(7)	F=1: -3155.69 F=2: -2589.08 F=3: -1742.32 F=4: -619.18 F=5: +775.31 F=6: +2434.84
$3d^24p\ ^4G^o_{7/2}$	147.2(3)	–	F=0: -2318.37 F=1: -2171.17 F=2: -1876.78 F=3: -1435.18 F=4: -846.39 F=5: -110.40 F=6: +772.79 F=7: +1803.18
$3d^24p\ ^4G^o_{9/2}$	83.3(6)	-25(4)	F=1: -1615.74 F=2: -1445.51 F=3: -1190.89 F=4: -852.78 F=5: -432.36 F=6: +68.88 F=7: +649.16 F=8: +1306.42
$3d^24p\ ^4G^o_{11/2}$	44.7(3)	-39(7)	F=2: -1032.47 F=3: -892.03 F=4: -706.22 F=5: -476.24 F=6: -203.63 F=7: +109.76 F=8: +461.81 F=9: +850.06
$3d4s4p\ ^2P^o_{1/2}$	240(19)	–	F=3: -539.63 F=4: +419.71
$3d4s4p\ ^2P^o_{3/2}$	327(4)	–	F=2: -2209.77 F=3: -1227.65 F=4: +81.84 F=5: +1718.71
$3d^24p\ ^4F^o_{3/2}$	280.3(3)	–	F=2: -1892.07 F=3: -1051.15 F=4: +70.08 F=5: +1471.61
$3d^24p\ ^4F^o_{5/2}$	121.1(3)	–	F=1: -1362.56 F=2: -1120.32 F=3: -756.98 F=4: -272.51 F=5: +333.07 F=6: +1059.77
$3d^24p\ ^4F^o_{7/2}$	67.5(3)	–	F=0: -1062.39 F=1: -994.94 F=2: -860.03 F=3: -657.67 F=4: -387.86 F=5: -50.59 F=6: +354.13 F=7: +826.30
$3d^24p\ ^4F^o_{9/2}$	44.4(3)	–	F=1: -854.11 F=2: -765.37 F=3: -632.26 F=4: -454.79 F=5: -232.94 F=6: +33.28 F=7: +343.86 F=8: +698.82
$3d^24p\ ^4D^o_{1/2}$	432.9(9)	–	F=3: -974.03 F=4: +757.58
$3d^24p\ ^4D^o_{3/2}$	126.2(15)	–	F=2: -851.93 F=3: -473.30 F=4: +31.55 F=5: +662.62
$3d^24p\ ^4D^o_{5/2}$	70.5(9)	–	F=1: -792.58 F=2: -651.67 F=3: -440.32 F=4: -158.51 F=5: +193.74 F=6: +616.45
$3d^24p\ ^4D^o_{7/2}$	42.0(3)	–	F=0: -661.04 F=1: -619.07 F=2: -535.13 F=3: -409.22 F=4: -241.33 F=5: -31.48 F=6: +220.35 F=7: +514.14
$3d^24p\ ^2G^o_{7/2}$	169.4(12)	–	F=0: -2667.78 F=1: -2498.39 F=2: -2159.63 F=3: -1651.48 F=4: -973.95 F=5: -127.04 F=6: +889.26 F=7: +2074.94
$3d^24p\ ^2G^o_{9/2}$	71(4)	–	F=1: -1361.96 F=2: -1220.45 F=3: -1008.20 F=4: -725.20 F=5: -371.44 F=6: +53.06 F=7: +548.32 F=8: +1114.33
$3d^24p\ ^2F^o_{5/2}$	226.9(18)	–	F=1: -2553.11 F=2: -2099.22 F=3: -1418.39 F=4: -510.62 F=5: +624.09 F=6: +1985.75
$3d^24p\ ^2F^o_{7/2}$	39.9(6)	–	F=0: -627.99 F=1: -588.12 F=2: -508.37 F=3: -388.76 F=4: -229.27 F=5: -29.90 F=6: +209.33 F=7: +488.44
$3d^24p\ ^2D^o_{3/2}$	236(3)	–	F=2: -1590.55 F=3: -883.64 F=4: +58.91 F=5: +1237.09
$3d^24p\ ^2D^o_{5/2}$	41.7(15)	–	F=1: -468.80 F=2: -385.46 F=3: -260.44 F=4: -93.76 F=5: +114.60 F=6: +364.62
$3d4s5s\ ^4D_{1/2}$	-739.6(24)	–	F=3: +1664.07 F=4: -1294.28
$3d4s5s\ ^4D_{5/2}$	595.1(6)	–	F=1: -6694.74 F=2: -5504.56 F=3: -3719.30 F=4: -1338.95 F=5: +1636.49 F=6: +5207.02
$3d4s5s\ ^4D_{7/2}$	603.8(12)	–	F=0: -9509.57 F=1: -8905.78 F=2: -7698.22 F=3: -5886.88 F=4: -3471.75 F=5: -452.84 F=6: +3169.86 F=7: +7396.33
$3d^24p\ ^4D_{3/2}$	56(4)	–	F=2: -376.39 F=3: -209.10 F=4: +13.94 F=5: +292.75
$3d^24p\ ^2D^o_{3/2}$	162.2(15)	–	F=2: -1094.77 F=3: -608.20 F=4: +40.55 F=5: +851.49
$3d^24p\ ^2D^o_{5/2}$	133.1(9)	–	F=1: -1497.46 F=2: -1231.25 F=3: -831.92 F=4: -299.49 F=5: +366.05 F=6: +1164.69
$3d^24p\ ^2P^o_{3/2}$	215.6(9)	–	F=2: -1454.97 F=3: -808.32 F=4: +53.89 F=5: +1131.64
$3d^24p\ ^2P^o_{1/2}$	110.3(15)	–	F=3: -248.23 F=4: +193.07
$3d4s4d\ ^4D_{5/2}$	440.4(18)	–	F=1: -4954.44 F=2: -4073.66 F=3: -2752.47 F=4: -990.89 F=5: +1211.09 F=6: +3853.46
$3d4s4d\ ^4D_{7/2}$	503.1(6)	–	F=0: -7923.06 F=1: -7420.01 F=2: -6413.91 F=3: -4904.75 F=4: -2892.55 F=5: -377.29 F=6: +2641.02 F=7: +6162.38
$(39861)_{5/2}$	-291.1(18)	–	F=1: +3274.86 F=2: +2692.66 F=3: +1819.37 F=4: +654.97 F=5: -800.52 F=6: -2547.11
$3d4s4d\ ^4G_{7/2}$	101.3(21)	–	F=0: -1595.94 F=1: -1494.62 F=2: -1291.96 F=3: -987.97 F=4: -582.65 F=5: -76.00 F=6: +531.98 F=7: +1241.29
$3d4s4d\ ^4F_{3/2}$	-215(10)	–	F=2: +1452.94 F=3: +807.19 F=4: -53.81 F=5: -1130.07
$3d4s4d\ ^4F_{5/2}$	188(4)	–	F=1: -2118.03 F=2: -1741.49 F=3: -1176.68 F=4: -423.61 F=5: +517.74 F=6: +1647.36
$3d4s5p\ ^2P^o_{3/2}$	136(4)	–	F=2: -918.71 F=3: -510.40 F=4: +34.03 F=5: +714.55
$3d4s4d\ ^4F_{7/2}$	279.4(15)	–	F=0: -4400.65 F=1: -4121.25 F=2: -3562.43 F=3: -2724.21 F=4: -1606.59 F=5: -209.56 F=6: +1466.88 F=7: +3422.73
$3d^24p\ ^2D^o_{3/2}$	-195(4)	–	F=2: +1317.36 F=3: +731.87 F=4: -48.79 F=5: -1024.62
$3d^24p\ ^2D^o_{5/2}$	469.2(18)	–	F=1: -5278.22 F=2: -4339.87 F=3: -2932.34 F=4: -1055.64 F=5: +1290.23 F=6: +4105.28
$3d4s5p\ ^2P^o_{3/2}$	758(9)	–	F=2: -5117.68 F=3: -2843.16 F=4: +189.54 F=5: +3980.42
$3d4s5p\ ^2F^o_{7/2}$	67(27)	–	F=0: -1057.67 F=1: -990.51 F=2: -856.21 F=3: -654.75 F=4: -386.13 F=5: -50.37 F=6: +352.56 F=7: +822.63
$3d4s5p\ ^2D^o_{3/2}$	342(4)	–	F=2: -2310.95 F=3: -1283.86 F=4: +85.59 F=5: +1797.41
$3d4s5p\ ^2D^o_{5/2}$	219(11)	–	F=1: -2468.79 F=2: -2029.89 F=3: -1371.55 F=4: -493.76 F=5: +603.48 F=6: +1920.17
$3d^24p\ ^2F^o_{5/2}$	11(5)	–	F=1: -128.16 F=2: -105.38 F=3: -71.20 F=4: -25.63 F=5: +31.33 F=6: +99.68
$3d^24p\ ^2P^o_{1/2}$	463(28)	–	F=3: -1040.80 F=4: +809.51

Measured A, B constants; sources in the data file.